

Sibtain Ahmed

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Education

Iowa State University [🔗](#)

Aug 2025 – Present | Ames, IA

PhD in Computer Science (MS expected Spring-2027)

Research Focus: Data Attribution for RL-Based LLM Fine-Tuning

- Investigating how individual training examples influence LLM behavior during reinforcement learning based fine-tuning, particularly in PPO/GRPO-style RLHF pipelines.
- Adapting efficient influence-function and gradient-similarity methods to score training samples by their estimated impact on validation loss and policy improvement.
- Developing data-selection and filtering strategies to remove harmful samples and improve RL fine-tuning stability, sample efficiency, and interpretability.
- Implementing experiments by modifying Hugging Face TRL PPOTrainer internals, including reward modeling, advantage estimation, KL-regularized policy updates, validation-gradient computation, and HPC-based GPU training.

NED University of Engineering & Technology [🔗](#)

Oct 2020 – Aug 2024 | Karachi, Pakistan

BE Computer Systems Engineering (CGPA: 3.61/4.00)

Professional Experience

Iowa State University [🔗](#)

Aug 2025 – Present | Ames, IA

Graduate Teaching Assistant – COMS 1270 & 3210

- Mentor 65+ students across Python programming, algorithms, problem-solving, and computer architecture, supporting labs, office hours, and coursework.
- Facilitate hands-on learning and provide individualized feedback on programming, assembly, CPU architecture, memory systems, and computer organization.

GoodCore [🔗](#)

Nov 2024 – Jun 2025 | Karachi, Pakistan

Software Engineer

- Led the development of BriefingHelper, an AI-powered data assistant integrated into BriefingSource [🔗](#), enabling non-technical users to query internal data in natural language, eliminating complex filters and manual SQL queries, cutting retrieval time from minutes to seconds.
- Built and deployed RESTful APIs using FastAPI, capable of handling 1,000+ concurrent users, and developed a Streamlit microservice leveraging LLM tool-calling, PgVector based retrieval (RAG), and PostgreSQL, improving query accuracy and reducing hallucination by ~50%.
- Developed a real-time tracking module to monitor token usage, API calls, and database queries, optimizing resource allocation and minimizing operating costs.
- Reduced LLM inference cost and latency by 40% by splitting a large model call into two optimized stages for SQL generation and response formatting.
- Containerized and deployed services via Docker, ensuring seamless integration and scalable operation within AWS infrastructure.
- Enhanced model reliability through prompt engineering, feedback-driven tuning, and continuous evaluation, driving adoption across multiple BriefingSource clients.

KoderLabs [🔗](#)

Aug 2023 – Oct 2023 | Karachi, Pakistan

Software Engineering Intern

- Contributed in integration of REST APIs for three Web and Android apps using React-Query, Next.js, React Native, and TypeScript, improving data flow consistency and UI responsiveness.
- Built state management, error-handling, and modular component logic, increasing client reliability and reducing load times by ~20%.
- Collaborated in Agile sprints with design, backend, and QA teams, engaging in pair programming and test-driven development to enhance code quality, version control practices, and cross-functional coordination.

Projects

EstatePermit [↗](#)

Jun 2026 – Jul 2026

AI-Powered Construction Permitting Platform

- Built PermitOS/EstatePermit, a multi-agent AI platform that automates construction permit review by coordinating specialized agents for jurisdictional, building-code, and site/environmental compliance analysis.
- Designed the agent orchestration layer using LangGraph, Band SDK, FastAPI, and LLMs, enabling agents to jointly parse project documents, flag compliance issues, and surface permitting requirements.
- Built document-driven workflows that convert raw project data into structured permit assessments, powering automated review, compliance reasoning, and permit-package generation.

RunwAI – Predictive AI for Air Traffic Safety [↗](#)

May 2026

AMD Developer Cloud Hackathon

- Built a multimodal AI pipeline fusing live flight (OpenSky), weather (METAR), and video streams via Qwen3-VL on AMD Instinct MI300X GPUs (ROCm/HIP), achieving sub-4ms detection latency at 10,000+ flight updates/sec
- Developed an explainable rerouting agent delivering confidence-scored, weather-aware flight path recommendations with >99% rules accuracy and <2s end-to-end response latency

Dialogify (FYP, Startup) [↗](#)

AI-powered Cloud Orchestration & Monitoring Solution

- Implemented an AI chatbot interface to automate AWS/GCP workflows, enabling non-technical users to deploy VMs through natural language prompts, reducing deployment time by 70% and steps from 11 to 4.
- Integrated Next.js frontend with optimized PostgreSQL backend, implementing tag-based IaC search that improved retrieval speed by ~50% and enhanced overall data handling efficiency.
- **Technologies:** Next.js, FastAPI, AssistantAPI, Zilliz Cloud, Postgres, Cloud Run (GCP)

AutoGrade [↗](#)

Nov 2025

Automated Grading of Scientific Notes | Agentic AI

- Built an agentic grading system for short scientific answers by modeling assessment as a semantic entailment and evidence-alignment problem.
- Benchmarked symbolic, transformer-based, and LLM-based approaches under low-resource and limited-supervision settings.
- Developed a robust NLP pipeline for handling paraphrasing, noisy student responses, and semantic variation in automated grading.

Achievements

- Section Leader – **Stanford Code-in-Place 2025**: Selected to lead and mentor global participants in Python programming and Problem-Solving fundamentals. [↗](#)
- Top 15% – **LeetCode** Contest Ranking: 8 Profile badges recognized for advanced problem-solving and algorithmic proficiency. [↗](#)
- **GRE** Quantitative Reasoning – Perfect Score (170/170, 94th Percentile Overall): Demonstrated exceptional quantitative skills. [↗](#)
- 95th Percentile – **National Standardized Quality Measurement Test**: Demonstrated exceptional analytical aptitude. [↗](#)
- **Harvard Puzzle Day 2025**: Solved 9/9 puzzles, ranked among top teams in logical reasoning & collaboration. [↗](#)

Volunteer and Teaching Experience

Co-founder – QubtiCoders [↗](#)

Nov 2023 – Jul 2024 | NED UET, Karachi

- Established a competitive programming community of 100+ members, promoting engagement in algorithms, contests, and coding challenges.
- Organized workshops, mentorship sessions, and coding competitions to strengthen peer learning and skill development.